

5(a)(i)	<i>any two from:</i> 1 (codon) 976 where, T is replaced by C / TCT becomes TCC or (codon) 977 where, A is replaced by T / TCA becomes TCT ; 2 ser(ine) in, both / cats and humans ; 3 <u>degenerate</u> code ;	2
5(a)(ii)	<i>any two from:</i> 1 base / C, deleted at, second codon / (codon) 975 ; 2 causes frameshift ; 3 next 9 codons changed / all codons afterwards are changed ;	2
5(a)(iii)	<i>any two from:</i> 1 <i>ref. to</i> shortened polypeptide ; 2 no tyrosinase made ; 3 change in, 3D / tertiary, shape / structure or change in active site ; 4 (so) tyrosinase, non-functional / cannot bind to substrate / inactive ; 5 tyrosine not converted to, DOPA / dopaquinone ;	2

Question	Answer	Marks
5(a)(iv)	<i>any one from:</i> <i>why</i> 1 large amount of, data / DNA sequences ; 2 fast / accurate / efficient ; A description of fast / accurate / efficient / detects each small difference = accurate <i>any one from:</i> <i>conclusions</i> 3 more similarity the more recent a common ancestor / AW ; 4 molecular clock ;	2
5(b)	<i>parent phenotypes:</i> (albino x fully pigmented) <i>parent genotypes:</i> t^at^a x Tt^s ; <i>gametes:</i> (t^a T t^s) <i>F1 genotypes:</i> Tt^a t^st^a ; <i>F1 phenotypes:</i> fully pigmented Siamese and <i>ratio:</i> 1 : 1 ; <i>F1 phenotypes must match genotypes</i> <i>ecf for mp2 and mp3 if parent genotype incorrect</i>	3